

Unstop and ET AI Hackathon 2.0 - Phase 2: Build Sprint - Prototype Report

Project Name: PoliticAI / ELECTOR Feed

Prepared by: Raghuvaran D

Position: Freelancer

Problem Statement: PS 6: AI for Digital Public Safety: Defeating Counterfeiting, Fraud & Digital Arrest Scams

Live Demo: https://politicai-rose.vercel.app/

GitHub Repository: https://github.com/RaghuvaranDBECse32/Politicai

Executive Summary

PoliticAI is a retrieval-augmented generation (RAG) political news and civic intelligence system designed to help users explore political developments, election-related reporting, misinformation trends, and region-specific public-interest issues using grounded evidence instead of unsupported generation. The system is built around source-linked retrieval, metadata-aware document organization, and a refusal-capable answer layer so that factual safety is preserved even when questions are ambiguous or evidence is incomplete. This makes the project suitable for a public-safety-oriented hackathon submission where trust, traceability, and reliability matter as much as usability.

The project is framed as a safety-first AI prototype for digital public safety because misinformation, fake claims, and misleading political narratives can cause real-world harm during elections and high-attention civic events. Rather than presenting itself as a general chatbot, PoliticAI acts as a controlled knowledge interface that retrieves evidence first, then generates bounded answers with citations. This design makes the system more defensible in evaluation because it clearly distinguishes between what can be supported and what should be refused.

Project Overview

The product direction combines political intelligence, civic literacy, and misinformation verification into a single interface. Users can search for political developments, ask verification-style questions, and review cited summaries of recent events. The application is designed to be inspectable, meaning that every factual answer is intended to be traceable to one or more retrieved sources.

The report should present PoliticAI as a structured AI system rather than a content generator. The core idea is that the knowledge source is the retrieved evidence, not the model memory. This distinction is important for the hackathon because it demonstrates careful system design, especially in high-risk public-information environments where hallucinations or unsupported statements can be harmful.

The project also has practical value because political and civic information is often fragmented across different sources, languages, and regions. A retrieval-based assistant can reduce the burden on users by collecting, filtering, and summarizing relevant information in one place. That makes the prototype more than a demo: it becomes a useful civic research layer.

Problem Statement

Political information environments are noisy, fast-changing, and highly vulnerable to misinformation. Users often encounter conflicting claims, incomplete reporting, or posts that appear credible but lack evidence. In such conditions, a standard generative AI assistant can easily produce outdated or fabricated answers if it is not constrained by retrieval and safety logic.

This challenge becomes more serious in multilingual and region-sensitive contexts, where local language reporting may be underrepresented and trust signals may be inconsistent. The result is that users may not only struggle to find accurate information, but also struggle to judge whether a claim is supported at all. PoliticAI is

designed to reduce that problem by requiring evidence before answer generation and by refusing questions that cannot be grounded.

The project also addresses the wider risk of unsupported civic summarization. A model that sounds fluent can still be wrong, and in political contexts that can create confusion or amplify misinformation. By making retrieval, citations, and refusal mandatory parts of the pipeline, the system treats safety as a product feature rather than an afterthought.

Objectives

The first objective is to build a queryable knowledge layer over political and civic reporting using structured, source-linked records. The second objective is to reduce hallucination by requiring retrieval evidence before generation. The third objective is to preserve citation and source traceability for every high-stakes response so that users can inspect where the answer came from.

A fourth objective is to support multilingual and region-sensitive use cases relevant to Asia and the broader Global South. A fifth objective is to demonstrate a practical AI safety MVP that is feasible within a hackathon timeline. These objectives help position the project as both technically meaningful and realistic to build.

Why This Matters in Asia

The Asia track is relevant because misinformation, governance complexity, and language diversity are especially important in this region. Many users need accessible civic information, trusted election references, and safer ways to examine political claims. A tool like PoliticAI can support those needs by making public-interest information easier to search and verify.

In India and neighboring contexts, election literacy and voter education are already recognized priorities. A civic intelligence system that draws from official references and curated reporting can complement these efforts by helping users find relevant information quickly. That makes the prototype socially relevant, not just technically interesting.

The regional fit is also important for judging. A solution that reflects local public-interest problems is usually stronger than a generic AI app because it shows awareness of context, harm, and deployability. PoliticAI can therefore be framed as a region-aware safety system for civic knowledge access.

Proposed Solution

The proposed solution is a safety-first RAG application that ingests curated political and civic sources, stores them with metadata, and retrieves evidence before generating an answer. The model is only used after relevant context has been gathered. This architecture reduces the risk of unsupported generation and improves answer traceability.

The system should support two broad flows. In the exploration flow, users ask about political developments, governance events, or misinformation narratives and receive grounded summaries. In the verification flow, users test a claim or ask for clarification, and the system either returns evidence-backed support or refuses the response if the evidence is too weak.

The solution should emphasize source ranking, recency, and region filtering. These controls matter because a political question is often only useful if the answer is not just fluent but also current, attributable, and contextually relevant. That is the core advantage of the proposed design.

System Architecture

The architecture can be explained in four layers: ingestion, knowledge storage, retrieval, and response. The ingestion layer collects trusted civic and political sources. The knowledge layer normalizes and tags documents by date, geography, language, source type, and topic so that later retrieval is safer and more precise.

The retrieval and reasoning layer searches the indexed corpus, reranks the results if needed, and passes only the evidence into the model prompt. The interface layer then presents answers with citations, timestamps, and refusal messages when grounding is weak. This clear separation between search, reasoning, and presentation is important because it shows a controlled AI workflow.

For the report, it helps to describe the architecture as evidence-first. The model should never be the sole source of truth. Instead, it should behave like an analyst that summarizes verified material and admits uncertainty when the retrieved evidence is insufficient.

Data and Source Strategy

The source strategy should begin with official civic and election resources, followed by curated reporting and clearly attributable public-interest content. The report should state that the MVP avoids uncontrolled scraping and instead prioritizes trusted references. This is a strong safety choice because evidence quality directly determines answer quality.

The system should also preserve source provenance. Every document should carry metadata such as source name, publication date, region, language, and topic. This makes the retrieval stage more controllable and helps users understand why a source appeared in the answer.

A good report should explain that source quality is not just a data issue but a safety issue. If the input corpus is weak, the output will be weak even if the model is strong. That principle should be made explicit in the submission.

RAG Pipeline Design

The pipeline can be described in eight steps: ingest, clean, chunk, tag, index, retrieve, decide, and generate. First, trusted political and civic content is collected. Second, it is cleaned and broken into manageable evidence units. Third, metadata is attached so later filtering is possible.

Next, the documents are indexed using lexical or hybrid search. After a user query is received, the system retrieves the most relevant chunks, optionally reranks them, and then decides whether the evidence is sufficient. Only if the grounding threshold is met does the system generate an answer from the retrieved context.

The report should also mention refusal logic. If no strong context match is found, the assistant should reject the request or ask for a narrower query rather than speculate. This is one of the strongest safety signals the project can demonstrate.

Safety and Hallucination Controls

The main safety principle is bounded generation. That means the assistant should not answer from memory alone and should not invent facts beyond the retrieved evidence. The system should also use strict system prompts, source-bounded answering, and citation requirements for every factual response.

Useful controls include refusal on weak context, query logging for later review, and separation of trusted civic knowledge from volatile claims. Optional multilingual expansion should preserve the same evidence-first logic. This makes the system safer across different use cases without changing the overall design philosophy.

The report should clearly state that RAG is not automatically safe. If retrieval quality is poor, hallucinations can still occur. That is why the project needs both retrieval evaluation and refusal thresholds, not just a powerful language model.

MVP Features

The MVP should include a political query interface, source-linked answer cards, region and topic filters, and an election/civic explainer mode. It should also include a fake-versus-real news verification flow, a refusal engine for unsupported questions, and session history for auditability.

If possible, the report should show one or two screen examples or describe the expected interface layout. Judges usually respond well when the user journey is easy to understand. The more clearly the prototype demonstrates safety and usability together, the stronger the submission becomes.

Evaluation Plan

The evaluation plan should prioritize groundedness over eloquence. Suggested test sets include in-scope factual queries with known evidence, paraphrases of the same queries, out-of-scope questions that should trigger refusal, and ambiguous questions with insufficient support. This shows that the system is being tested under realistic conditions.

Useful metrics include grounded answer rate, refusal precision, unsupported-claim rate, citation correctness, retrieval hit rate, and latency per query. If multilingual capability is part of the prototype, add multilingual retrieval success rate. These metrics give the evaluation section a stronger engineering focus.

The report should explain that the goal is not only to produce good answers, but to avoid bad ones. In safety-oriented systems, good refusal behavior can be as important as answer quality. That is an important message for judges to see.

Expected Outcomes

If implemented well, the project should show that a political AI assistant can be made substantially safer by prioritizing retrieval, attribution, and refusal. It should also show that civic and political information tools for Asia can be built with stronger local relevance and safer operating boundaries. That makes the project meaningful both technically and socially.

The expected outcome section should avoid overclaiming. Instead, it should say that the MVP demonstrates a workable architecture, a safety-first answer policy, and a foundation for future expansion. This gives the submission a professional and realistic tone.

Limitations

The report should openly state that political news changes quickly, so freshness is always a challenge. Retrieval quality also depends on how well the dataset is curated and tagged. Multilingual performance may vary if local-language sources are sparse.

Bias can still enter through source selection, even in a grounded system. The model is also safer for summarization and lookup than for predicting motive, intent, or future outcomes. Transparent limitations make the project look more credible, not weaker.

Future Scope

Future work can include Indic-language support, better contradiction handling, stronger source ranking, misinformation case tracking, and human review workflows for disputed claims. These additions would make the system more useful and more robust over time.

The future scope should preserve the same principle: retrieval-first, source-linked, refusal-capable political intelligence rather than unconstrained generative politics. That consistency is important because it shows the design is guided by safety, not only by features.

Conclusion

PoliticAI is a strong hackathon concept because it addresses a real and difficult deployment problem: how to help users explore political information without encouraging hallucination, bias amplification, or unsupported claims. By combining structured source ingestion, retrieval-first answer generation, and refusal when evidence is weak, the project offers a practical and regionally relevant AI safety MVP.

For the final report, the writing should be expanded with examples, subsection detail, and a clear explanation of the user flow so the document comfortably reaches around ten pages. The best approach is to add more depth in the architecture, evaluation, and limitations sections rather than adding filler. That keeps the report long enough for submission while still sounding technical and polished.

Visual Direction for Report

Suggested visual for cover: professional illustration of a civic intelligence dashboard with fact-check badges, source citations, shield icons, and political news cards.

Suggested visual for architecture section: layered diagram showing ingest -> chunk -> index -> retrieve -> verify -> cite -> refuse.

Suggested visual for safety section: AI moderation screen with warning, refuse, and citation states.

Suggested visual for features section: dashboard mockup showing query, retrieved evidence, and cited answer cards.

Design Notes

Use a clean professional theme with blue, gray, and white colors.

Add one cover image, one architecture diagram, and one safety-flow illustration.

Keep screenshots and icons minimal and aligned with a formal report style.

Use consistent fonts, headings, and spacing so the document feels like an academic or enterprise submission.